

Rakta Daan

Answers to the lecture questions on my app and App Store Optimization (ASO)

1. What is my app about?

In Nepal, people sometimes die while their family is running around looking for blood, even though that blood is sitting in a blood bank or a hospital only a few kilometres away. The problem is not always that there is no blood. Most of the time nobody can see where it is, and nothing connects the patient to it fast enough.

Rakta Daan is a Flutter mobile app that I am building to fix this. When someone needs blood, the app looks in three places, in this order: **blood banks first, then hospitals, then donors nearby**, using the phone's GPS. The idea is to find a match within minutes. Getting the blood to the patient is handled by our NGO and blood bank partners, so the family does not have to go and fetch it themselves.

Donors get something back for helping. The plan includes food, coupons, free blood test reports, health suggestions from AI, a lifetime record of every donation, public recognition, and a 5% discount if the donor ever needs blood. Local businesses can advertise through those rewards, NGOs can run campaigns and apply for government grants, and hospitals pay a monthly software fee. That way the app can pay for itself while still helping people.

One thing I want to be careful about is the money side. Donated blood should stay free. Blood banks in Nepal charge a service fee for testing and handling, so patients would only pay that normal fee (plus delivery), not a price for the blood itself. The app is still in the building stage and has not been released yet.

2. How big is my app?

The slide says an app should ideally be no bigger than about **150 MB on iOS** and about **100 MB on Android**. My target is to stay far below both. A Flutter app with maps, login and lists of blood banks should land somewhere around 30 to 50 MB, but that is only a guess until I make a real release build and measure it.

Size matters a lot for this app. Many phones in Nepal have limited storage, mobile data is not cheap, and internet in some areas is slow. A person who needs blood urgently will not wait for a huge download, and someone with a full phone will delete a big app first. A small app also gets installed more often, which helps ASO later.

What I plan to do to keep it small:

- Compress all images and use vector icons (SVG) wherever possible.
- Avoid large videos, big fonts and unused packages in the project.
- Publish Android as an app bundle so each phone only downloads what it needs.
- Build with the `--split-per-abi` option when I share test APKs.
- Check what is taking space with `flutter build apk --analyze-size`.
- Load extras such as maps or reports only when the user opens them.

3. What is ASO?

ASO stands for App Store Optimization. It means improving how your app appears in the Google Play Store or the Apple App Store so that more people find it and install it. The slide mentions two places where this matters: **search results** and **top ranking charts**. Search results are what appear when a user types something like "blood bank". Top charts are the lists such as Top Free or Top Trending.

The main things you can improve are:

- **Title:** the strongest place for keywords. For example "Rakta Daan - Find Blood Fast".
- **Keywords:** the words people really search, like blood donor, blood bank near me, emergency blood, and Raktadan.
- **Description:** explains what the app does. It needs to be clear in the first two lines.
- **Icon and screenshots:** most people decide in a few seconds by looking at these.
- **Ratings and reviews:** both the number of reviews and the average stars.
- **Downloads and retention:** the stores notice if people install the app and keep using it.

4. How is ASO connected to SEO?

They are very close. SEO (Search Engine Optimization) is about getting a website to show up high on Google. ASO does the same job inside an app store. In both cases the goal is to be found without paying for adverts, and in both cases keywords are the starting point. The difference is mainly where the ranking happens and what the ranking looks at.

	SEO	ASO
Where it ranks	Google, Bing and other search engines	Google Play Store and Apple App Store
What is ranked	Web pages	Apps
Keywords	In page titles, headings and content	In app title, subtitle or short description, and keyword fields
Trust signals	Backlinks from other sites, page speed	Ratings, reviews, downloads, low crash rate
Goal	More visitors to a website	More installs of an app

For Rakta Daan there is one more link. I will also need a small website so that people searching on Google find the app, and that website can point to the store page. So SEO and ASO will work together.

5. Why do ASO?

The slide gives three reasons, and here is what each one means for my app.

Increase app downloads

Most people do not install an app because they saw an advert. They open the store and search. If Rakta Daan does not appear when someone types "blood donor" or "blood bank", it might as well not exist. Good ASO brings in free, organic installs, which is important because a new project like mine has almost no money for paid ads.

Brand exposure

Every time the app shows up in search results or a chart, people see the name and the logo. Even if they do not install it that day, they remember it. For an emergency app this is very important, because when a family suddenly needs blood they should already know the name Rakta Daan.

Increase reviews and ratings

Stars and reviews affect both ranking and trust. A patient's family, a donor or a hospital manager will look at the rating before deciding to install. More good reviews push the app higher, and a higher position brings in more users, so it keeps feeding itself.

6. iOS or Android?

The slide says it depends on which one you are going for, because the two stores work differently even though they share some similarities.

	Google Play (Android)	Apple App Store (iOS)
Where keywords are read	Title, short description and the full long description	Title, subtitle and a hidden keyword field. The description is not searched
Rough limits	Title 30 characters, short description 80, full description 4000	Title 30 characters, subtitle 30, keyword field 100
Changing the listing	Usually quicker to update	Normally changed together with a new app version that Apple reviews
Similar in both	Ratings, reviews, downloads, retention, crash rate, and a clear icon and screenshots matter on both stores.	

Because Android is the more common phone in Nepal, I plan to launch on **Android first** and add iOS later. That also means I will write the Play Store description carefully, since Google reads all of it for keywords.

7. My ASO plan for Rakta Daan

Here is how I would apply all of this once the app is ready:

App title and short description

Title: **Rakta Daan - Find Blood Fast**

Short description: *Find blood banks, hospitals and donors near you in minutes.*

Keywords to use

blood donor, blood bank, blood bank near me, emergency blood, need blood, donate blood, Raktadan, Rakta Daan, Nepal blood, hospital blood

Screenshots (in this order)

- The search screen showing blood banks, hospitals and donors on a map.
- A blood request being matched, with the time it took.
- The donor rewards page.
- The donor's lifetime record and health report.
- Partner NGOs and delivery status.

Getting good reviews

I will not ask for a rating in the middle of an emergency, because that would be wrong. The best time is after something has gone well, for example after a request is completed or after a donor finishes a donation. A short, polite pop-up at that point should get honest and positive reviews. I will also reply to every review, especially the bad ones.

Checking the results

After launch I will track how many people see the listing, how many install, and which keywords bring them in. If a keyword does not work I will replace it and test again.

Summary

Rakta Daan is meant to connect patients with blood that already exists nearby. Keeping the app small makes it usable on cheap phones. ASO is how people will find it in the store without paid adverts, and because Android and iOS work differently, I will start with Android and build the listing around clear keywords, good screenshots and honest reviews.